AUE-CSC 104: COMPUTER HARDWARE

Course Units: 3 Units (C: LH 30; PH 45)

MODULE 1: COMPUTER CIRCUITS

1.1 Introduction to Computer Circuits

Computer circuits form the physical foundation of computer systems. They consist of interconnected electronic components designed to process, store, and transmit electrical signals that represent data and instructions.

1.2 Basic Electronic Components

- **Resistors:** Limit and control the flow of electric current.
- **Capacitors:** Store and release electrical energy; used for filtering and timing.
- **Inductors:** Store energy in magnetic fields; used in power regulation.
- **Transistors:** Act as electronic switches or amplifiers and are the building blocks of digital circuits.

1.3 Diodes and Diode Arrays

- **Diodes:** Semiconductor devices that allow current to flow in only one direction.
- **Functions of Diodes:** Rectification, voltage regulation, and signal clipping.
- **Diode Arrays:** Multiple diodes integrated into a single package for compact and efficient circuit design.

1.4 Peripheral Interface Adapters (PIAs)

PIAs enable communication between the CPU and peripheral devices. They manage data transfer, control signals, and synchronization between hardware components.

MODULE 2: INTEGRATED CIRCUITS (ICs)

2.1 Meaning of Integrated Circuits

An integrated circuit (IC) is a complete electronic circuit fabricated on a small semiconductor material, usually silicon.

2.2 IC Fabrication Process

1. Silicon crystal growth
2. Wafer slicing and polishing
3. Oxidation

4. Photolithography
5. Etching
6. Doping
7. Metallization
8. Packaging and testing

2.3 Levels of Integration

- **SSI (Small Scale Integration):** Logic gates and flip-flops
- **MSI (Medium Scale Integration):** Adders, counters, multiplexers
- **LSI (Large Scale Integration):** Microprocessors and memory chips
- **VLSI (Very Large Scale Integration):** Modern CPUs and SoCs

2.4 Hardware Design Using MSI, LSI, and VLSI

These technologies enable compact design, increased speed, reduced power consumption, and improved reliability in computer hardware.

MODULE 3: MEMORY SYSTEMS

3.1 Overview of Computer Memory

Memory stores data and instructions required by the computer system for processing.

3.2 Primary Memory

- **RAM (Random Access Memory):** Volatile memory used for active processes
- **ROM (Read Only Memory):** Stores permanent instructions such as firmware

3.3 Secondary Memory

Non-volatile storage used for long-term data retention.

3.4 Core Memory

Core memory uses magnetic cores to store binary data. It was widely used in early computers and is non-volatile but bulky and expensive.

MODULE 4: MAGNETIC STORAGE DEVICES

4.1 Magnetic Disks

- Hard Disk Drives (HDDs)
- High-capacity random access storage

4.2 Magnetic Tapes

- Sequential access storage
- Used for backups and archival purposes

4.3 Video Disks

- Optical and magnetic storage technologies
- Examples: CDs, DVDs, Blu-ray disks

MODULE 5: PERIPHERAL DEVICES

5.1 Input Devices

- **Keyboards:** Alphanumeric data entry
- **Character Recognition Devices:** OCR and MICR systems

5.2 Output Devices

- **Printers:** Dot matrix, inkjet, laser
- **CRTs (Cathode Ray Tubes):** Early visual display units

5.3 Importance of Peripherals

They enable interaction between users and the computer system.

MODULE 6: OPERATIONAL AMPLIFIERS (OP-AMPS)

6.1 Definition

Operational amplifiers are high-gain electronic amplifiers used in analog circuits.

6.2 Characteristics of Op-Amps

- High input impedance
- Low output impedance
- High voltage gain

6.3 Applications

- Signal amplification
- Filtering
- Voltage regulation

MODULE 7: ANALOG-TO-DIGITAL AND DIGITAL-TO-ANALOG CONVERTERS

7.1 Analog-to-Digital Converter (ADC)

Converts continuous analog signals into discrete digital values.

7.2 Digital-to-Analog Converter (DAC)

Converts digital data into analog signals for output devices.

MODULE 8: ANALOG COMPUTERS

8.1 Definition

Analog computers operate on continuous data represented by physical quantities such as voltage or current.

8.2 Characteristics

- High-speed computation
- Continuous data processing
- Limited accuracy

8.3 Applications

- Engineering simulations
- Control systems

RECOMMENDED TEXTBOOKS

- M. Morris Mano – *Computer System Architecture*
- Andrew S. Tanenbaum – *Structured Computer Organization*
- Thomas L. Floyd – *Electronic Devices*